



AI Traffic Signal System

Project Brief — Roman Urdu Mein Mukhtasar Wazahat

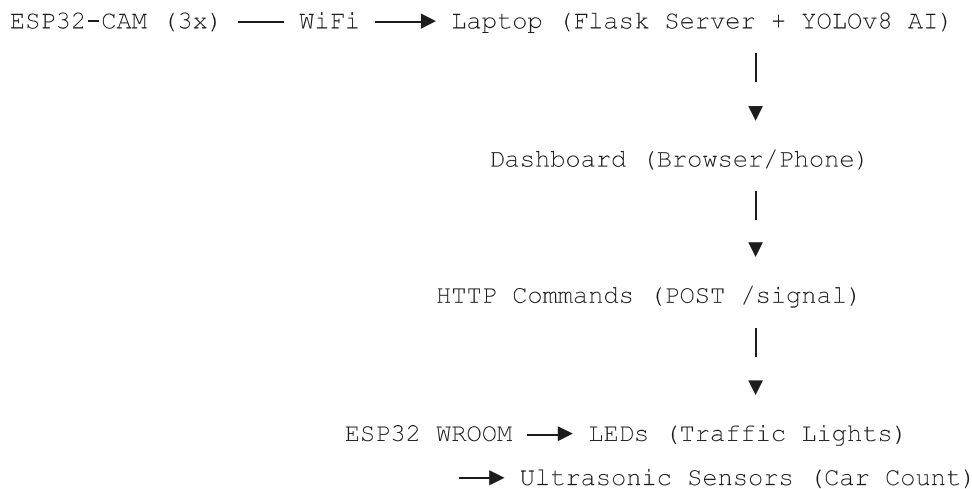
Prepared: June 2026 | Version: 1.0 | ESP32 + YOLOv8 + Flask

1. Project Ka Overview

Yeh ek **AI-based smart traffic light system** hai jo 3 roads (lanes) ke traffic ko automatically control karta hai. Iss system mein **cameras** lagaye gaye hain jo gaariyon ko detect karti hain, **ultrasonic sensors** hain jo gaariyan count karti hain, aur ek **web dashboard** hai jis se sab kuch phone ya laptop se control hota hai.

Ek line mein: Camera se traffic dekhna → AI se gaariyan count karna → us hisaab se green light ka time set karna — sab automatic aur remotely phone se bhi control.

2. System Kaise Kaam Karta Hai — Full Flow



1. **ESP32-CAM** (3 cameras) har lane ka live video stream karti hain WiFi ke through.
2. **Laptop (Flask Server)** un streams ko receive karta hai aur **YOLOv8 AI model** se gaariyan detect karta hai.
3. AI bata'ta hai ke lane mein **1 gaari hai, 2 hain, ya 3 hain** — us hisaab se green light ka time change hota hai.
4. **Dashboard** browser mein open hota hai — laptop pe ya phone pe — jahan se sab control hota hai.
5. Dashboard se jo bhi command aaye, wo **HTTP request** ke through ESP32 WROOM ko bhejta hai.
6. **ESP32 WROOM** un commands ko receive karke physically **LEDs on/off** karta hai.

3. Hardware — Kya Kya Lagaya Gaya Hai

Component	Quantity	Kaam Kya Hai
ESP32 WROOM (Dev Module)	1	Master controller — sab LEDs aur sensors isi se connected hain
ESP32-CAM (AI Thinker)	3	Cameras — har lane ke liye 1, live video stream karti hain
Traffic LED Module (R/Y/G)	3	Red, Yellow, Green lights — har lane ke liye 1 set
Ultrasonic Sensor (HC-SR04)	3	Gaariyan count karta hai — har lane ke liye 1
Breadboard + Jumper Wires	—	Wiring connections ke liye

4. Software — Konsi Technologies Use Ho Rahi Hain

Technology	Kahan Use Ho Raha Hai
Python (Flask)	Main server — backend logic, API endpoints, sab kuch yahan handle hota hai
YOLOv8 (Ultralytics)	AI model — camera se aaye frame mein gaariyan detect karta hai
OpenCV	Video frames decode karna aur AI ke liye prepare karna
HTML/CSS/JavaScript	Dashboard UI — browser mein dikhne wala interface
Arduino C++ (ESP32)	Firmware — ESP32 boards pe chalti hai, WiFi + LEDs + sensors control
mDNS	Hostnames (traffic-esp.local, cam-lane1.local) — IP yaad rakhne ki zaroorat nahi

5. WiFi Connection — Kaise Connect Hota Hai

Sab devices ek hi WiFi network pe hain:

WiFi SSID: TrafficSys | **Password:** System@123

- ESP32 WROOM power on hote hi automatically WiFi se connect hota hai.
- Teenon ESP32-CAMs bhi usi WiFi se connect hoti hain.
- Laptop pe Flask server chalta hai, jo sabko connect karta hai.
- Agar WiFi disconnect hojaye, toh ESP32 **khud reconnect** try karta hai har 5 second mein.
- **mDNS** use hota hai, yani IP address yaad rakhne ki zaroorat nahi — fixed hostnames hain (jaise traffic-esp.local).

6. Modes — System Ke 3 Modes

● A) Sequence Mode (Automatic)

Yeh default mode hai. Isme traffic light khud cycle karta hai:

Lane 1 GREEN → Lane 2 GREEN → Lane 3 GREEN → Lane 1 GREEN → ...

Har lane ka cycle:

Phase	Duration	Kya Hota Hai
● Yellow (pre-green)	2 second	Warning — green aane wali hai
● Green	5s (default) / 7s / 10s	Gaariyan guzarti hain — time AI decide karta hai
● Yellow (post-green)	2 second	Warning — red aane wali hai
● Red (clearance)	3 second	Sab red — safe transition

● B) Manual Mode

Isme automatic cycling ruk jaata hai. User **khud choose karta hai** ke kis lane ko GREEN, YELLOW ya RED karna hai. Dashboard pe har lane ke liye 3 buttons hain — GREEN, YELLOW, RED. Phone se bhi control ho sakta hai.

● C) Emergency Mode

Isme do options hain:

- **Stop All (All Red):** Teenon lanes RED hojati hain — sab traffic ruk jaata hai.
- **Open Lane X:** Sirf ek specific lane GREEN hoti hai, baaki sab RED — jaise ambulance ke liye raasta kholna.

Emergency clear karne ke liye "Clear Emergency" button hai. AI bhi **automatically emergency trigger** kar sakta hai agar **accident detect** ho (sab RED kar deta hai).

7. AI Detection — Gaariyan Kaise Detect Hoti Hain

Har lane ke liye ek ESP32-CAM hai jo live video stream karti hai. Flask server us video ko receive karta hai aur **YOLOv8 AI model** se analyse karta hai.

AI Kya Detect Karta Hai (4 Classes):

Detection	Matlab	Green Light Time
OneCar	Lane mein 1 gaari hai	5 second
TwoCar	Lane mein 2 gaariyan hain	7 second
ThreeCar	Lane mein 3 ya zyada gaariyan hain	10 second
Accident	Accident detect hua!	Sab RED — Emergency

Matlab: Jis lane mein zyada traffic hai, uski green light zyada der tak rahegi. Kam traffic wali lane ki green light kam time hogi. Yeh sab **AI automatically decide** karta hai.

AI Model Kaise Bana:

- **Roboflow** pe 221 images ka dataset banaya (toy cars ki images).
- YOLOv8 format mein annotate kiya (OneCar, TwoCar, ThreeCar, Accident).
- Model train karke best.pt file banayi — yeh AI ka "brain" hai.

8. Ultrasonic Sensors — Gaariyan Kaise Count Hoti Hain

Har lane mein ek **HC-SR04 ultrasonic sensor** lagaya gaya hai. Yeh sensor sound waves bhejta hai aur measure karta hai ke koi cheez kitni door hai.

- Agar koi object **5 cm se kam** distance pe aaye → **1 gaari count** hoti hai.
- Jab gaari nikal jaaye (distance wapas barh jaaye) → sensor ready for next car.
- Double-counting se bachne ke liye **state tracking** hai — ek gaari sirf 1 baar count hogi.
- Dashboard pe har lane ka count dikhayi deta hai aur "Reset All" button se zero ho sakta hai.

Backup ke tor pe: Agar cameras offline hon, toh sensor counts se bhi green light ka time decide hota hai (≥ 4 cars = 10s, ≥ 1 car = 7s).

9. Dashboard — Phone/Laptop Se Control

Flask server `http://localhost:5000` pe chalta hai. Phone se bhi access hota hai (`http://<laptop-IP>:5000`), bas same WiFi pe hona chahiye.

Dashboard Ke Buttons Aur Features:

Button / Feature	Kya Karta Hai
Connect / Connect All	ESP32 aur cameras ko WiFi se connect karta hai
System ON/OFF	Poora system on/off karta hai — OFF hone pe sab RED
Sequence / Manual toggle	Automatic ya manual mode mein switch karta hai
GREEN / YELLOW / RED buttons	Manual mode mein har lane ka signal change karta hai
Default / Equal / Custom Timing	Green light ka default time set karta hai
Stop All (All Red)	Emergency — sab lanes RED
Open Lane 1/2/3	Emergency — sirf ek lane GREEN, baaki RED
+5s to Current Green	Current green lane ka time 5 second barha deta hai
Clear Emergency	Emergency khatam karke normal mode pe wapas
Reset All Counters	Ultrasonic sensor ke counts zero karta hai
Camera Set / Stop	Har lane ki camera ka stream start/stop karta hai
Live Video Feeds	Teenon cameras ka live view with AI annotations
Traffic Light Animation	Har lane ki light ka real-time status dikhata hai

Keyboard Shortcuts (Laptop pe):

Key	Kaam	Key	Kaam
1	Lane 1 GREEN	A	Sequence Mode
2	Lane 2 GREEN	M	Manual Mode
3	Lane 3 GREEN	Esc	Clear Emergency

10. ESP32 WROOM — Wiring (Pin Connections)

Traffic LEDs:

Lane	● Red Pin	● Yellow Pin	● Green Pin
Lane 1	GPIO 23	GPIO 22	GPIO 21
Lane 2	GPIO 19	GPIO 18	GPIO 5
Lane 3	GPIO 26	GPIO 25	GPIO 4

Ultrasonic Sensors:

Lane	TRIG Pin	ECHO Pin	Power
Lane 1	GPIO 13	GPIO 12	5V (VIN)
Lane 2	GPIO 14	GPIO 27	5V (VIN)
Lane 3	GPIO 15	GPIO 32	5V (VIN)

11. Project Files — Konsi File Kya Karti Hai

File	Kya Karti Hai
app.py	Main server — sab backend logic, AI, API endpoints yahan hain
templates/index.html	Dashboard ka poora UI — HTML + CSS + JavaScript sab ek file mein
models/best.pt	Trained YOLOv8 AI model — gaariyan detect karne ka "brain"
firmware/traffic_controller/traffic_controller.ino	ESP32 WROOM ka code — LEDs control + Ultrasonic sensors
firmware/CameraWebServer/CameraWebServer.ino	ESP32-CAM ka code — video stream + mDNS hostname
firmware/wifi_config.h	WiFi SSID aur password — sab boards isi se padhte hain
data/ folder	AI training dataset — 221 annotated images (Roboflow se)

12. System Chalane Ka Tareeqa

1. **WiFi Hotspot ON karein** — SSID: TrafficSys, Password: System@123
2. **ESP32 WROOM ko USB se connect karein** — automatically WiFi se connect hoga aur sab LEDs RED hojaengi.
3. **Teenon ESP32-CAMs ko power dein** — har ek WiFi se connect hogi.
4. **Laptop pe command chalayein:** python app.py
5. **Browser mein kholein:** http://localhost:5000
6. **"Connect All" button dabayein** — sab devices connect hojaenge.
7. System automatic sequence mode mein chalna shuru ho jaayega!

Phone se access: Phone ko same WiFi pe connect karein aur browser mein http://<laptop-ka-IP>:5000 kholein.

13. Features Ka Khulaasa

#	Feature	Status	Kaise Kaam Karta Hai
1	WiFi Connection	☒ Done	ESP32 auto-connect + mDNS hostnames
2	Manual Mode	☒ Done	Dashboard se har lane ka signal manually set
3	Emergency Alert	☒ Done	Stop All / Open Lane / +5s — phone se control
4	Sequence Timing (5s, 2s, 3s)	☒ Done	Green=5s, Yellow=2s, Red clearance=3s cycle
5	Phone Se Control	☒ Done	Same WiFi pe phone browser se poora control
6	Traffic Density Based Timing	☒ Done	AI detect karta hai: 1 car=5s, 2=7s, 3=10s
7	Cloud Server (Remote Access)	☒ Nahi	Abhi sirf local WiFi pe kaam karta hai
8	Ultrasonic Sensor Count	☒ Done	HC-SR04 se gaariyan count + dashboard pe display